Survival Generative Adversarial Network: Applications on the MSK-50K cohort

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1 Introduction

Survival analysis characterizes the distribution of time-to-event outcomes such as death, relapse, or treatment failure under right-censoring, and underlies decision making across many biomedical fields. The growing scale of clinical and genomic registries, together with stringent privacy constraints and the need to augment small or unbalanced cohorts, has motivated the development of generative models that produce synthetic survival data while preserving the joint distribution of covariates, event times, and censoring indicators.

Generative adversarial networks (GANs) achieve good performance on synthetic tabular data, but the time-to-event setting introduces difficulties absent from the standard problem. First, outcomes are right-censored: the observed pair $(t_i, e_i)_{i=1}^n$, time and censorship, is only a partial summary of the latent event time, and the generator must respect the conditional structure that censoring imposes. Second, the joint (x, t, e) distribution, where x is the covariate, is constrained by survival-specific properties such as a monotone survival function and a censoring mechanism. Third, downstream evaluation must rely on survival-aware metrics rather than marginal fidelity alone. Direct GAN generation of (x, t, e) tuples tends to produce optimistic survival times, attenuated tails, and broken covariate–outcome dependencies [Norcliffe et al., 2023].

SurvivalGAN [Norcliffe et al., 2023] addresses these issues through a decoupled three-model architecture: a conditional WGAN with gradient penalty generates covariates (x) conditional on event status and a Bayesian Gaussian mixture cluster label, a DeepHit network Lee et al. [2018] maps generated covariates to a discrete-time survival function (S), and an XGBoost regressor Chen and Guestrin [2016] maps covariates and survival predictions to event times (t). Such decoupling makes survival structure explicit rather than implicit in the GAN, and the original paper reports several metrics specifically on survival data generation, including *optimism*, *short-sightedness*, and *Kaplan–Meier divergence*.

In this work we replicate and extend SurvivalGAN on the MSK-IMPACT 50K cohort [Bandlamudi et al., 2026], a pan-cancer registry roughly an order of magnitude larger than some of the benchmark data sets considered in the original paper. Our results are threefold. (i) We describe a reproducible preprocessing pipeline that resolves identifier redundancy, missing or implausible survival labels, multi-sample patients, and high-cardinality categorical variables in the raw archive. (ii) We report a hyperparameter configuration that scales SurvivalGAN to a substantially larger and more heterogeneous cohort, with a deeper time-to-event regressor and an enlarged generator. (iii) We benchmark SurvivalGAN against CTGAN Xu et al. [2019], a strong tabular-only baseline, along marginal, joint,

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survival-distributional, and downstream-utility, and quantify where survival-aware modeling delivers genuine gains over a generic tabular GAN.

2 The MSK-50K data set

The MSK-50K data set [Bandlamudi et al., 2026] collects the overall survival of cancer patients together with their demographic, histopathological, and genomic sequencing data. The release used in this work consists of 54331 tumour samples drawn from 48179 unique patients across 64 cancer types. For every sample the archive records 45 raw fields, spanning sample identifiers, demographic variables (age, sex, genetic ancestry), high-level and fine-grained cancer labels, genome-wide summary statistics from FACETS (ploidy, fraction of genome altered, purity, whole-genome doubling), immunogenetic typing (six HLA class-I alleles with per-allele LOH status), mutational-burden indices (`Mutation Count`, `TMB Score`, `MSI Score` and `MSI Type`), clinical annotations (`Disease Status`, `Metastatic Site`, `Sample Type`), and the survival outcome `Overall Survival (Months)` together with `Overall Survival Status`.

The generative model requires a covariate vector paired with a scalar event time T and binary event indicator $\delta \in \{0, 1\}$. The raw archive cannot be used directly because (i) several columns are identifiers; (ii) a non-trivial fraction of rows carry missing or invalid survival outcomes; (iii) some patients contribute multiple tumour samples and would be over-represented; and (iv) a number of fields have cardinality comparable to the sample size itself, carrying negligible signal while inflating the one-hot representation produced by the tabular encoder (e.g., a categorical variable with 100 levels). We therefore apply the preprocessing pipeline summarized in Figure 1.

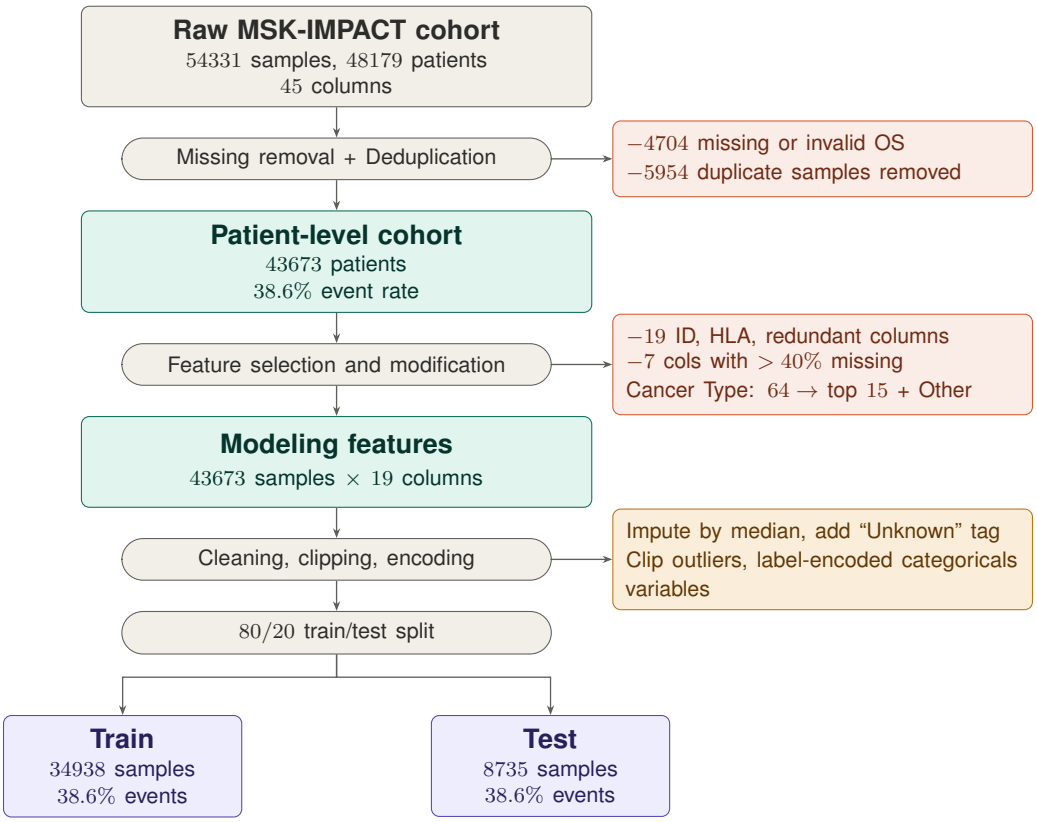

Figure 1: Preprocessing pipeline for the MSK-IMPACT 50K cohort.

2.1 Data preprocessing

We coerce `Overall Survival (Months)` to a numeric event time T and map `Overall Survival Status` to a binary event indicator δ, with $\delta = 0$ denoting a living (right-censored) subject and $\delta = 1$

denoting a deceased subject. Samples with missing survival labels or with non-positive or implausibly large survival times ($T \leq 0$ or $T \geq 600$ months) are excluded, removing 4704 tumours. Because some patients contribute multiple tumour samples, we then deduplicate to one sample per patient, retaining the primary sample where available and otherwise the first sample encountered. This step removes a further 5954 secondary samples and yields a patient-level cohort of 43673 subjects with an event rate of 38.6% (16866 deaths, 26807 censored).

We retain 17 covariates from the 45 raw columns: ten continuous variables (`Age at Diagnosis`, `Ploidy (FACETS)`, `FACETS Estimated Purity`, `Fraction Genome Altered`, `MSI Score`, `Mutation Count`, `Number of Other Cancer Types`, `Sample coverage`, `TMB Score`, `Pathologist Estimated Tumor Purity`) and seven categorical variables (`Sex`, `Cancer Type`, `Disease Status`, `MSI Type`, `Genetic Ancestry`, `FACETS QC`, `Whole Genome Doubling Status`).

Remaining missingness is resolved by simple imputation: continuous columns are imputed at the column median, and categorical columns at value `Unknown`. To prevent extreme outliers by the GAN's tabular encoder, we clip several important variables to be coherent to their physical meanings (e.g., `Fraction Genome Altered` should be bounded by $0, 1$).

The resulting analysis cohort contains 43673 patients and 19 columns.

3 The SurvGAN model

The SurvGAN is a three-model architecture specialized to train and generate synthetic survival data. It specifically contains a Wasserstein-GAN (WGAN), a deepHIT model (deep neural network for predicting hitting times) and an extreme gradient boosting model (XGB). To explain the architecture, we require some notations. Let $(x_i, t_i, e_i)_{i=1}^n$ be the observed data, where $x \in^p$ represents covariates, $t > 0$ represents observed survival time and $e \in \{0, 1\}$ is the censorship indicator.

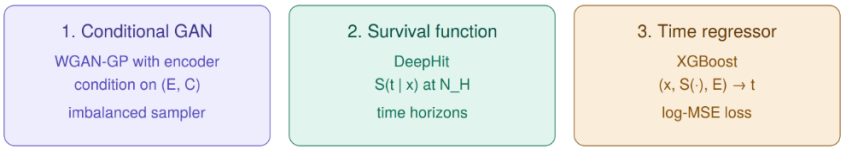

Figure 2: The training workflow of survGAN. WGAN-GP: Wasserstein GAN with gradient penality. DeepHit: Deep neural network for estimating hitting probabilities.

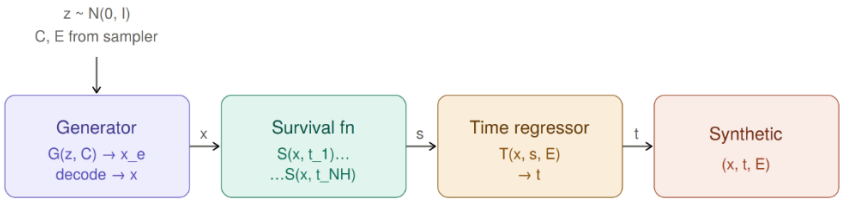

Figure 3: The generating workflow of survGAN.

The training process is summarized in Figure 2. Data are sampled from the MSK data conditioning on event e and a class encoder C, where $C = f(x)$ is generated by a Bayesian Gaussian Mixture Model. Then, a WGAN is trained on the data set. At the same time, a deepHIT model and an XGB model are trained using the MSK data. The deepHIT model takes x as input and, conditioning on e, generates the estimated survival probability $\hat{S}(r \mid x) := \hat{P}(t > r \mid x)$. The XGB model takes $x, \hat{S}, e$ as input and generates the estimated survival time, t. After the training process of these three models, synthetic data of any length can be generated by following the workflow in Figure 3.

3.1 Model parameters

In Table 1, we report the hyper-parameters used to train the survGAN model, along side with the default values reported in Norcliffe et al. [2023]. In this work, we increase the size of XGB model and the generator model. We also make tweaks on the model achitecture of discriminator to see the impact on the performance.

Table 1: SurvivalGAN hyperparameters used in our MSK-IMPACT experiments compared with the values reported by Norcliffe et al. (2023). ✓ = match, ≈ = close, × = differs.

Component	Parameter	Our value	Paper value	Status
Survival function (DeepHit)	Model	DeepHit	DeepHit	✓
	No. durations	100	100	✓
	Batch size	100	100	✓
	No. epochs	2000	2000	✓
	Learning rate	10^{-3}	10^{-3}	✓
	Hidden width	300	300	✓
	α	0.28	0.28	✓
	σ	0.38	0.38	✓
	Dropout rate	0.02	0.02	✓
	Patience	20	20	✓
	Batch normalisation	True	True	✓
Time-to-event regressor (XGBoost)	Model	XGBRegressor	XGBRegressor	✓
	No. estimators	500	200	×
	Max depth	6	5	×
	Booster	gbtree (default)	Dart	×
	Tree method	default	Histogram	×
Conditional WGAN-GP	Sampling strategy	`imbalanced_time_censoring`	With data freq.	×
	No. iterations	3000	1500	×
	Gen. hidden layers	3	3	×
	Gen. hidden width	256	250	×
	Gen. non-linearity	Tahn	Tanh	×
	Gen. dropout	0.0	0.1	×
	Disc. hidden layers	2	2	✓
	Disc. hidden width	256	250	≈
	Disc. non-linearity	Leaky ReLU	Leaky ReLU	✓
	Disc. dropout	0.1	0.1	✓
	Learning rate	10^{-3}	10^{-3}	✓
	Weight decay (G / D)	10^{-4} / 10^{-5}	10^{-3}	×
	Batch size	256	500	×
	Gradient penalty λ	10	10	✓
	Encoder max clusters	5	10	×

4 Results

We evaluated SurvivalGAN trained on the MSK data set along the three axes laid out in our proposal: marginal covariate fidelity, joint covariate structure, and survival-specific quality. All evaluations compare 2,500 synthetic samples against held-out real samples of the same size.

From Figure 4, we see that the synthetic frequencies are essentially indistinguishable from real for many discrete variables, with per-level differences within 1–2 percentage points. This confirms that the conditional GAN's encoder/decoder pipeline preserves discrete structure faithfully even when category counts are highly imbalanced. Continuous variables are well captured. However, many variables suffer from a multi-mode problem. This is primarily due to the Bayesian GMM class encoder, setting too many possible modes for the covariate.

To evaluate whether the synthetic samples preserve joint structure, we projected real and synthetic covariates jointly into 2D via t-SNE (t-distributed Stochastic Neighbor Embedding, Figure 5). t-SNE is a machine learning algorithm used for dimensionality reduction, particularly for visualizing high-dimensional data in 2D spaces. The synthetic points are interleaved throughout every cluster occupied by the real points, with no cluster left empty and no synthetic-only artifact clusters appearing in

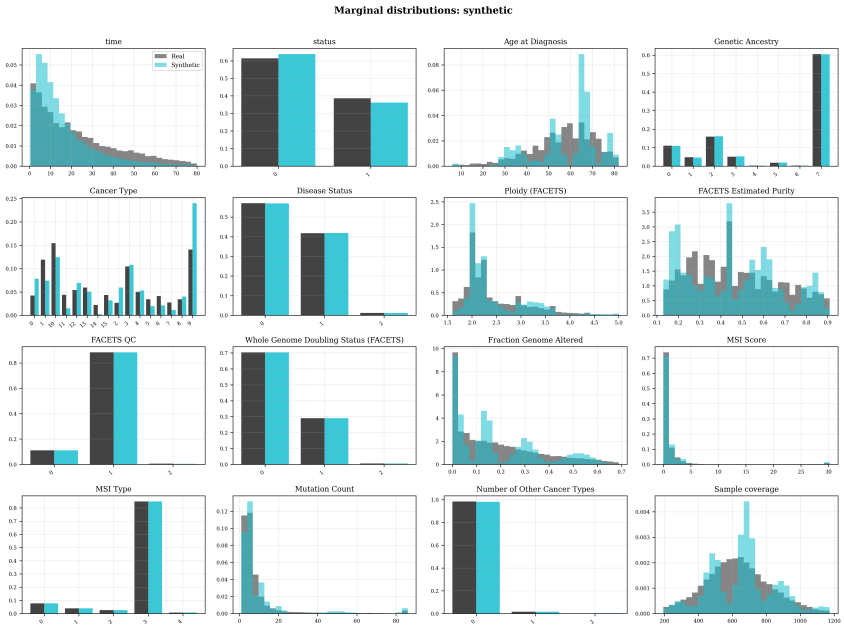

Figure 4: Side-by-side histograms of real (**dark**) and synthetic (**cyan**) values for sixteen representative features spanning continuous, ordinal, and categorical types

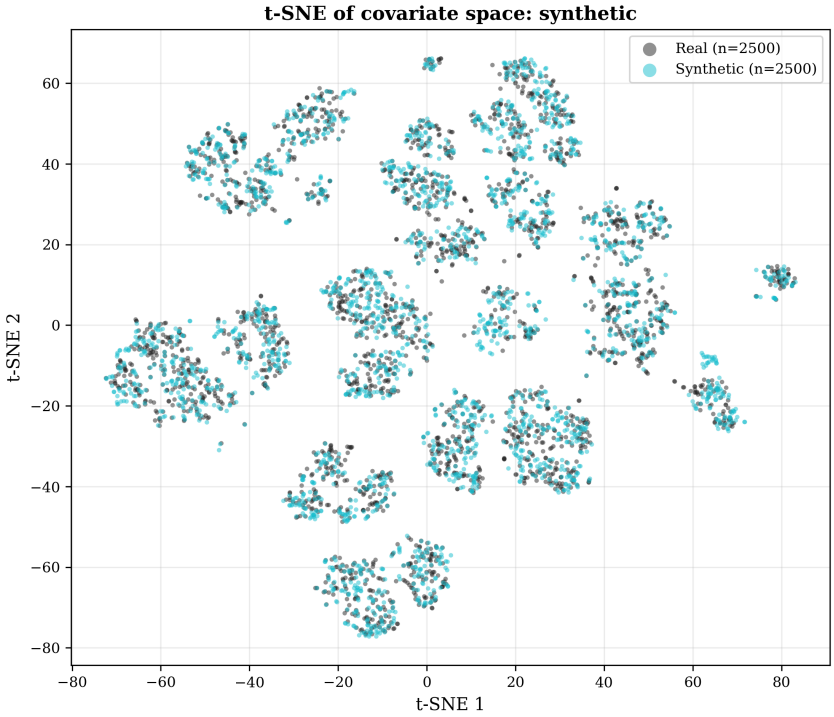

Figure 5: t-distributed Stochastic Neighbor Embedding for MSK data set: real vs synthetic.

Table 2: Synthetic-data quality comparison between SurvivalGAN and CTGAN on MSK-IMPACT 50K. Panels (a)–(b) report distribution-level metrics computed against the held-out real training and test splits; panel (c) reports downstream survival utility on the real test set. Within each row, **bold** marks the better value. TRTR = Train Real / Test Real (identical baseline for both methods); TSTR = Train Synthetic / Test Real.

Metric	**SurvivalGAN**		**CTGAN**	
	train	test	train	test
(a) Covariate similarity				
Jensen–Shannon (marg. mean)	**0.006**	**0.006**	0.012	0.012
Wasserstein dist. (joint)	**0.096**	**0.096**	0.165	0.163
α-precision (OC)	**0.926**	**0.922**	0.763	0.718
β-recall (OC)	**0.442**	**0.462**	0.381	0.369
Authenticity (OC)	0.524	0.513	**0.565**	**0.557**
Detection AUC (MLP)	**0.520**	0.496	0.603	**0.501**
(b) Survival distribution similarity				
Optimism	-0.036	-0.036	$\mathbf{+0.027}$	$\mathbf{+0.027}$
Kaplan–Meier divergence	**0.046**	**0.042**	0.089	0.091
Short-sightedness	**0.000**	**0.000**	0.067	0.058
Censoring-rate difference	**0.012**	**0.009**	0.135	0.131
(c) Downstream survival utility (Harrell's C on real test)				
	TRTR	TSTR	TRTR	TSTR
Cox proportional hazards	0.630	0.621	0.630	0.621
XGBoost (AFT)	0.605	**0.628**	0.605	0.539
Feature-importance Spearman (ρ, XGB)	**0.733**		0.027	

regions of low real density. This is a qualitative evidence that the WGAN is generating data that captures the joint covariate distribution of the real data set.

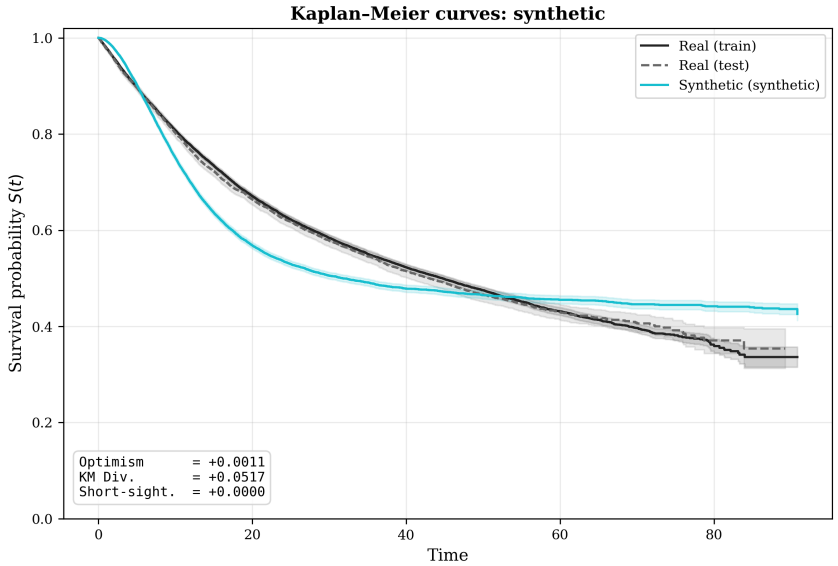

Figure 6: Kaplan–Meier curves estimated from the real training split (**solid black**), the held-out real test split (dashed gray), and the synthetic samples (cyan).

The synthetic curve in Figure 6 sits slightly below both real curves through the middle of the time range, then converges back to the real curves.

Table 2 aggregates the comparison against CTGAN along three axes. SurvivalGAN dominates on every survival-distribution metri, indicating that the censoring mechanism is well calibrated only when the generator is explicitly informed about event status. Similar for covariate balancing. The

one metric on which CTGAN is competitive is authenticity (0.557 vs. 0.513), which we attribute to the WGAN's tendency to interpolate between training points more aggressively than CTGAN's mode-specific normalization. The Cox concordance trained on SurvivalGAN's synthetic data is within 0.01 of the train-real / test-real ceiling, and the Spearman rank correlation between feature importances obtained from real and synthetic training data jumps from $\rho = 0.03$ (CTGAN) to $\rho = 0.73$ (SurvivalGAN). Taken together, these results indicate that SurvivalGAN preserves marginal fidelity and the conditional structure that downstream survival learners should exploit.

5 Conclusion

We replicated SurvivalGAN on the MSK-IMPACT 50K cohort and benchmarked it against CTGAN. Several findings are interesting. First, every survival-distribution metric is better for SurvivalGAN than for the tabular-only CTGAN baseline, and the feature-importance correlation between real- and synthetic-trained learners rises. This supports the original argument of Norcliffe et al. [2023] that survival structure should be modelled explicitly outside the GAN rather than implicitly through joint (x, t, e) generation. Second, marginal histograms show spurious multi-modality in a subset of continuous covariates, which we trace to the Bayesian Gaussian mixture class encoder activating too many mixture components on a heterogeneous pan-cancer cohort; reducing the maximum cluster count from 10 to 5 mitigated but did not eliminate the artefact. Finally, β-recall remains at roughly 0.45 against α-precision of 0.92, indicating that the generator covers high-density regions well but under-samples the tails.

Several extensions are natural. Conditional generation on cancer type or genetic ancestry could be used to oversample under-represented strata for downstream survival benchmarks, addressing a real limitation of the raw MSK cohort. The current pipeline also discards HLA loss-of-heterozygosity status and several high-cardinality genomic variables; replacing label encoding with a learned categorical embedding, or substituting the WGAN backbone with a diffusion model for tabular data, may further improve both marginal fidelity and tail coverage. We leave these directions to future work.

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A Appendix